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Methane-rich gas emissions from natural geologic seeps can be chemically distinguished from anthropogenic leaks

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Fossil methane, a key component of the atmospheric greenhouse gas budget, is emitted by fugitive leaks from petroleum industry activities and natural geologic seepage. Since the gas from these two sources can exhibit similar isotopic compositions, differentiating them using atmospheric observations is often challenging. Here, we provide a conceptual model that helps distinguish the two sources. Using molecular and isotopic data from global seep and reservoir gas inventories, and new seep analyses from major hydrocarbon-bearing basins in California, we identify six post-genetic changes in gas chemistry. These changes include alterations of molecular ratios among alkanes (ethane, propane, and butane) and ^{13}C enrichment of propane and carbon dioxide due to oxidation and secondary methanogenesis. Such changes are typical of seepage, and do not occur in deep reservoir gas that may leak during extraction and transport. Our model can aid in source attribution of fossil gas analyzed in airborne or field surveys and serve as a foundation for future work on regional emissions of natural gas.

Fossil methane (CH_4), the radiocarbon-free hydrocarbon produced by ancient thermogenic or microbial processes in sedimentary rocks and used as an energy resource, is an important component of the atmospheric greenhouse gas budget contributing to climate changes[1–3]. It is released to the atmosphere through the Earth's natural degassing (i.e., geologic seepage), with an estimated global emission of ~45 [27–63] Tg CH_4 yr^{-1}[2,4] (see also ref. 5 and ref. 6 for debates and uncertainties regarding global emission estimates), and through anthropogenic, fugitive emissions from oil–gas industry activities, including petroleum production, transport, and refining, with a global emission estimate of ~91 [73–118] Tg yr^{-1}[2]. Thermogenic gas accumulations account for approximately 80% of the world's natural gas resources[7]; consequently, the majority of fossil methane emissions are thermogenic[8].

Fossil gas emitting sites are assessed by means of ground-based measurements and by satellite and aircraft observations[9–14]. Yet, if the emitting source is unknown, it can be challenging to differentiate fugitive hydrocarbon leaks from natural seepage because both frequently have the same origin (i.e., the underlying gas reservoir). As a result, their molecular and isotopic compositions may overlap. In particular, the stable C isotopic composition of CH_4 ($\delta^{13}C$–C_1) of natural seepage and underground reservoirs can be isotopically indistinguishable[15–17]. Consequently, certain atmospheric observations centered on fugitive emissions may inadvertently include gas from natural seepage, which is common in many oil and gas production regions, and can contribute to elevated methane in groundwater, the soil, and the atmosphere[10,18,19]. There are also many instances of ground-based measurements where it is unclear whether a surface gas manifestation (a bubbling pool, a flame from the soil) represents a natural seep or a well or pipeline leak (stray gas)[20–22]. In addition, in ecosystems with methanogenic activity (e.g., wetlands, rice paddies, and lakes), natural seeps may be erroneously interpreted as modern microbial gas emissions[23,24].

Although gas emissions from oil and gas infrastructure leaks and natural seepage commonly originate from the same underlying reservoir, they differ in their pathways to the surface. Leaking gas from oil and gas infrastructure follows direct and relatively rapid transport pathways from the reservoir to the surface (i.e., oil and gas wellbores and underground pipelines). Accordingly, fugitive emissions typically exhibit an indistinguishable molecular and isotopic composition from reservoir gas[25]. Natural seepage, by contrast, often follows more circuitous natural pathways, such as faults or fractures, to move from the reservoir to the surface. This longer subsurface residence time, and potential for greater interaction with groundwater, can lead to more extensive chemical alterations.

Here, we show that natural hydrocarbon seeps can actually exhibit diagnostic chemical (molecular and isotopic) characteristics that can be applied to discriminate natural vs anthropogenic hydrocarbon emissions for

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both ground-based and atmospheric measurements. We used global datasets of seeps and reservoir gas geochemistry (CH_4 and CO_2 isotopic composition, and methane to ethane + propane ratio, from 238 thermogenic natural seep samples and 5421 conventional thermogenic reservoir gas samples; see "Data availability") to evaluate the widespread dimension of the geochemical alterations due to seepage. For a detailed investigation exploring additional seepage proxies scarcely reported in the global datasets (e.g., the relative abundance of alkanes up to C_{6+}, stable C isotopic composition of ethane ($\delta^{13}C$–C_2) and propane ($\delta^{13}C$–C_3)), new data were obtained from 18 surface free-phase gas manifestations located in five hydrocarbon-bearing basins in California, USA, which are representative of the region's extensive petroleum resources (Fig. 1 and Supplementary Fig. 1). The investigated surface gas manifestations are predominantly natural seeps documented in the historical literature that pre-date local oil and gas extraction activities (e.g., La Brea tar pits), or for which there is no nearby anthropogenic source (e.g., an oil or gas well). Utilizing this comprehensive dataset, we show that natural emissions can frequently be distinguished from fugitive anthropogenic emissions by identifying an ensemble of post-genetic changes in thermogenic gas composition that occur during natural migration from reservoir to surface. Understanding these compositional changes may allow for improved identification of both natural seepage and fugitive leak contributions to atmospheric emissions in ground-based and remote sampling.

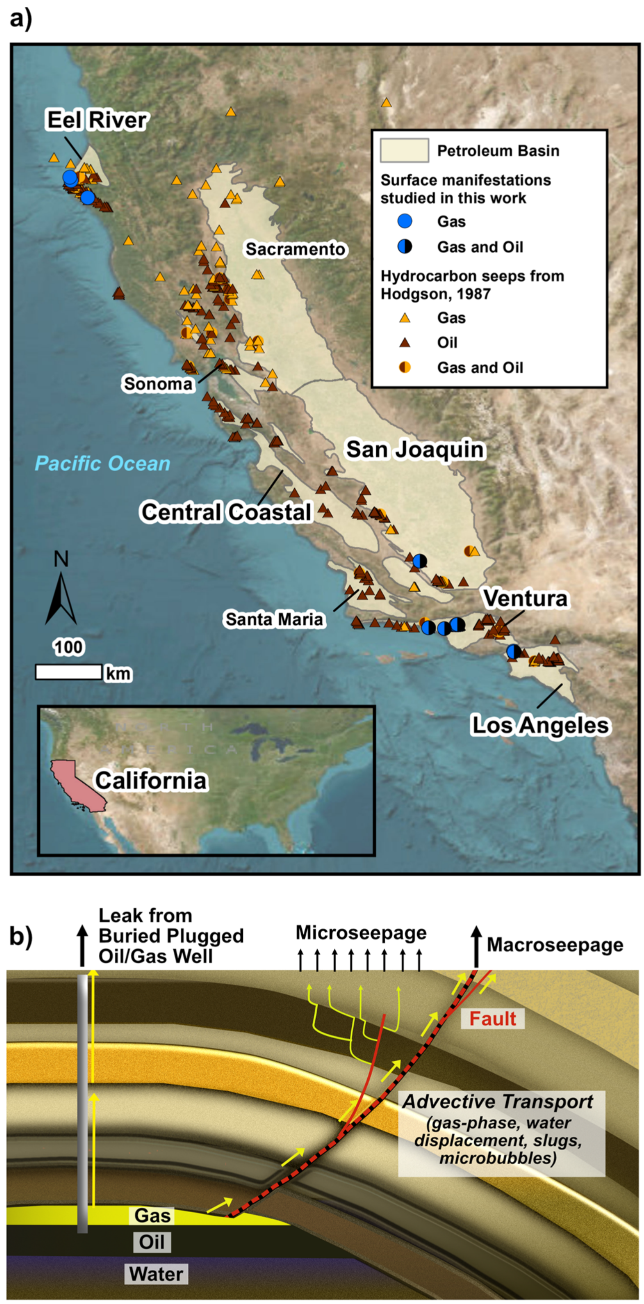

Fig. 1 | Location of surface gas manifestations in California and schematic representation of natural seepage and anthropogenic leakage. a Map showing the 18 surface gas manifestations investigated in this work, as well as the locations of published seepage data[30,44,59–62]; (**b**) Schematic representation of natural geologic seepage vs leakage from a buried plugged oil and gas well. Petroleum basins are from the California Geological Survey[70] and aerial imagery is provided by Esri[71].

Results

Global datasets of natural seep and reservoir gas geochemistry

The global datasets of thermogenic natural gas seepage (238 samples) and conventional thermogenic reservoir gas (5421 samples) display a similar distribution of $\delta^{13}C$–C_1 values (Fig. 2a). Such a similarity confirms the lack of isotopic fractionation during advective migration, as indicated in previous works[15–17,26,27]. However, there are notable differences in the distributions of $C_1/(C_2+C_3)$ ratios and $\delta^{13}C$–CO_2 values for these datasets (Fig. 2b, c, respectively). Specifically, 85% of reservoir gas samples exhibit a $C_1/(C_2+C_3)$ ratio less than 100, as compared to only 25% of seepage samples. Rather, seepage samples exhibit a broad range of $C_1/(C_2+C_3)$ ratios, with more than a third (38%) displaying ratios greater than 1000. Reservoir gas samples also exhibit a narrower range of $\delta^{13}C$–CO_2 values than seepage samples, with 92% of reservoir samples displaying $\delta^{13}C$–CO values between −20 and +5‰. By comparison, seepage gas samples exhibit a much wider distribution of $\delta^{13}C$–CO_2 values, with over half (51%) of seepage samples displaying positive $\delta^{13}C$–CO_2 values greater than +5‰.

These data point to a widespread difference between seeps vs reservoir gas (equivalent to fugitive emissions) geochemistry. The increased C_1/C_2+C_3 ratio and the ^{13}C-enrichment in CO_2 of seeps, mainly due to molecular fractionation during migration and secondary methanogenesis (as explained in more detail below), were already recognized for some regions, such as in the seepage systems of Azerbaijan, Trinidad, Italy, and Romania[15,16,28].

Seep and reservoir gas geochemistry in California

The molecular and isotopic compositions of CH_4 and CO_2 for all surface gas manifestations investigated in the five basins in California (Supplementary Fig. 1) are consistent with thermogenic origin, in some cases likely admixed with microbial gas (Fig. 3). Specifically, on the updated genetic diagrams of methane, based on more than 20,0000 data globally[29], i.e., $\delta^{13}C$–C_1 vs $C_1/(C_2+C_3)$ (Fig. 3a) and $\delta^{13}C$–C_1 vs δ^2H–C_1 (Fig. 3b), the samples predominantly plot in the characteristic oil-associated thermogenic gas field, and for a subset of samples, within early-mature and late-mature thermogenic gas fields. Approximately half the surface manifestations exhibit a methane contribution from secondary methanogenesis, a typical post-genetic process related to petroleum biodegradation observed in many seeps worldwide[28]. This conclusion is supported by the presence of positive $\delta^{13}C$–CO_2 values (Fig. 3c). One sample from the Salmon Creek surface manifestation exhibited particularly ^{13}C-enriched CH_4, but a δ^2H–C_1 value outside the thermogenic zonation of Milkov and Etiope[29] (Fig. 3b). The lower δ^2H–C_1 value could reflect a higher interaction with groundwater (partial equilibration with water's hydrogen) likely due to longer residence time in aquifers. This aspect will be discussed in more detail later as it can be diagnostic of a natural seepage process. In sum, all gas manifestations in this dataset derive from reservoirs of natural gas generated via catagenesis in mature source rocks, which represent the dominant geological source of hydrocarbons in California[30–32].

To better understand the chemical and isotopic characteristics of natural gas seeps, we compared compositional and isotopic data from the surface gas manifestations to available gas data from local reservoirs (i.e., see "Data availability"). Collectively, data from both surface gas manifestations

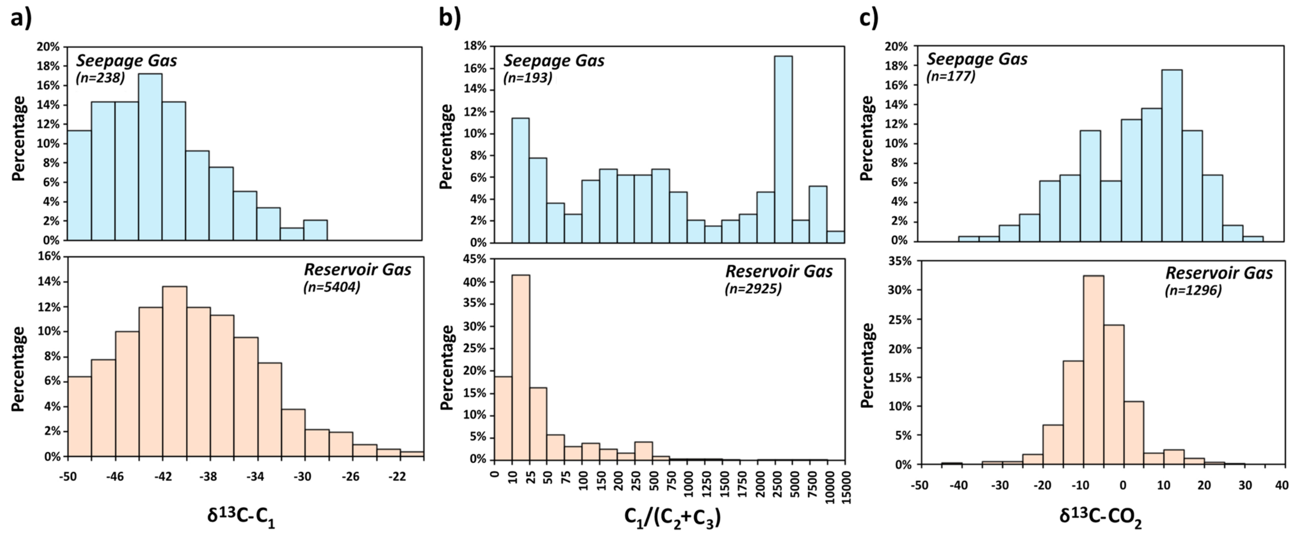

Fig. 2 | Distribution of stable C isotope ratio of methane, C_1–C_3 composition, and stable C isotope ratio of carbon dioxide for global datasets of thermogenic natural seepage gas vs conventional thermogenic reservoir gas. a $\delta^{13}C$–C_1; **b** $C_1/(C_2 + C_3)$; **c** $\delta^{13}C$–CO_2, where C_1: methane; C_2: ethane; and C_3 propane. The source of the data is reported in "Data availability".

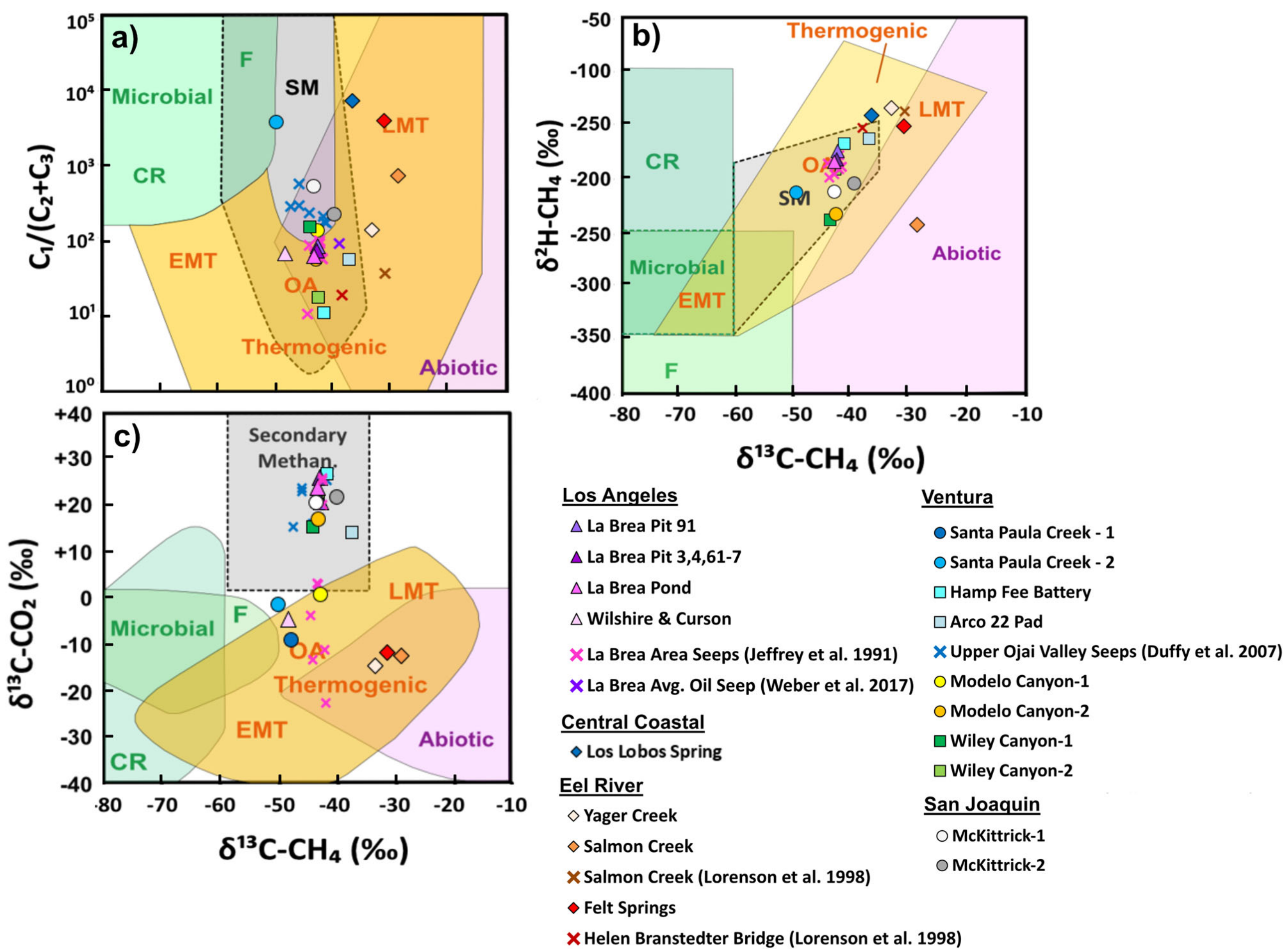

Fig. 3 | The surface gas manifestations in California within the natural gas genetic diagrams. a $C_1/(C_2 + C_3)$ vs $\delta^{13}C$–C_1; **b** δ^2H–C_1 vs $\delta^{13}C$–C_1; and **c** $\delta^{13}C$–CO_2 vs $\delta^{13}C$–C_1. Additional published data are shown as an x symbol. Genetic plots from Milkov and Etiope[29]. CR = CO_2 reduction, F = fermentation, SM = secondary methanogenic, EMT = early mature thermogenic, LMT = late mature thermogenic, OA = oil associated.

and local hydrocarbon reservoirs were available in the La Brea, Upper Ojai Valley, Wiley Canyon, San Ardo, and Eel River areas.

The majority (>90%) of surface gas manifestations exhibit lower quantities of C_2 and/or C_3, denoted by a higher $C_1/(C_2 + C_3)$ ratio, compared to the underlying reservoir. This finding is compatible with the global pattern discussed above. A few (<10%), however, exhibit $C_1/(C_2 + C_3)$ ratios approaching that of local reservoir gas. For example, in the La Brea area, the gas released at the intersection of Wilshire Boulevard and Curson Avenue[8] exhibits a $C_1/(C_2 + C_3)$ ratio only slightly higher than that of the local reservoir (Salt Lake field; Figs. 4a and 5a). This manifestation is characterized by particularly high gas emissions, prompting the city authorities to install a permanent passive ventilation system in 1999. Similarly, the Hamp Fee Battery (Upper Ojai Valley; Figs. 4b and 5c) and Wiley Canyon-2 (Wiley Canyon; Fig. 5b) are characterized by $C_1/(C_2 + C_3)$ ratios approaching those of local reservoir gases, as well as high gas fluxes, inferred from instantaneous exceedances of the methane lower explosive limit (LEL) 15 cm above the point of surface manifestation.

The stable C isotope composition ($\delta^{13}C–C_1$) of surface manifestations is not markedly different (i.e., typically <5‰) from that of reservoir gas (Fig. 4), as observed for the global pattern. However, the stable H isotope composition ($\delta^2H–C_1$), $\delta^{13}C–CO_2$, $\delta^{13}C–C_3$, C_2/C_3, and iC_4/nC_4 ratios of surface manifestations can exhibit substantial differences vs gas from local reservoirs. For example, in the Upper Ojai and San Ardo areas, three of the four surface manifestations with $\delta^2H–C_1$ data were characterized by >60‰ $\delta^2H–C_1$ enrichment compared to the local reservoir (Supplementary Fig. 3). More than half of the sampled surface manifestations contain $CO_2 > 10$ vol.%, and virtually all of these are characterized by $\delta^{13}C–CO_2 > +10‰$ (Supplementary Fig. 5). Nevertheless, within local seepage areas (e.g., La Brea), surface manifestations may exhibit a wide spread of $\delta^{13}C–CO_2$ values, e.g., from +27‰ to −25‰, though the $\delta^{13}C–C_1$ composition changes very little. Several surface manifestations exhibit higher C_2/C_3 and iC_4/nC_4 ratios relative to local reservoir gases. As an example, detected C_2/C_3 and iC_4/nC_4 ratios of La Brea manifestations are as high as 6.2 and 23 respectively, which are markedly higher than the same ratios in the reservoir gas (2.7 and 0.6, respectively). $\delta^{13}C$ of C_2 and C_3 data were only available for reservoir gases from the La Brea and Upper Ojai Valley; these data were compared to that of local surface manifestations (Supplementary Fig. 6). The majority of the surface manifestations in the La Brea area exhibited similar $\delta^{13}C$ of C_2, and C_3 (i.e., within ±5‰) to that of local reservoir gases. However, within the Upper Ojai Valley, several surface manifestations exhibited markedly enriched $\delta^{13}C–C_3$ (i.e., >+25‰) relative to reservoir gas.

Discussion

Conceptual model of chemical changes during thermogenic natural gas seepage

Combining the results of our analyses and literature data, we have developed a conceptual model describing molecular and isotopic alterations that are specific to gas seepage, and that can aid in the differentiation of natural seeps from anthropogenic leaks of thermogenic gas. As described below, we identified six primary post-genetic changes in gas chemistry during the natural migration of thermogenic reservoir gases to the surface (Fig. 6). Observations of these alterations in data suggest contributions from natural

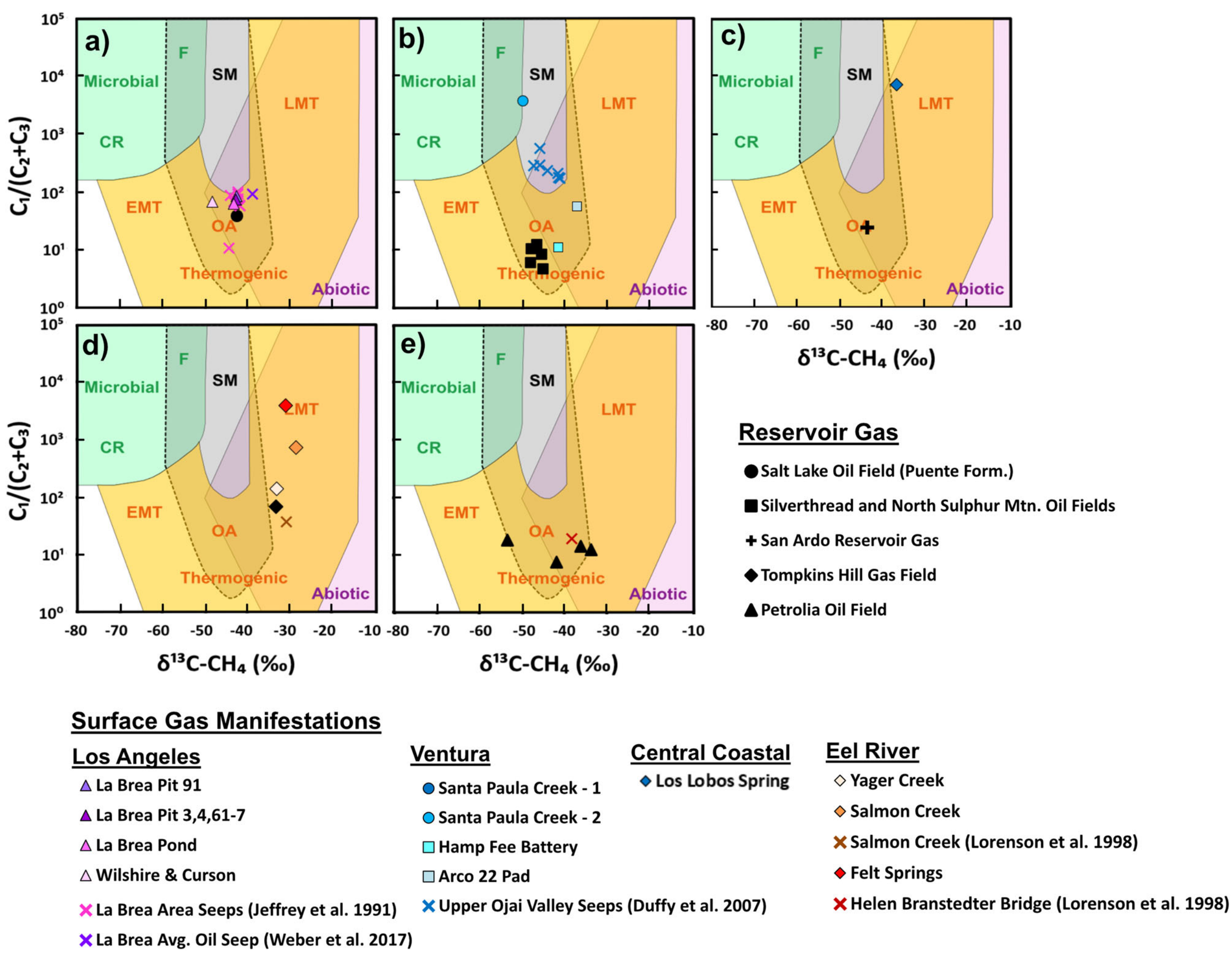

Fig. 4 | Comparison between surface manifestations and local reservoirs in terms of C_1–C_3 composition and stable C isotope ratio of methane. a The La Brea area; **b** the upper Ojai Valley; **c** the San Ardo oil field area; **d** the North Eel River Basin; and **e** the South Eel River Basin. Genetic diagrams as in Fig. 3.

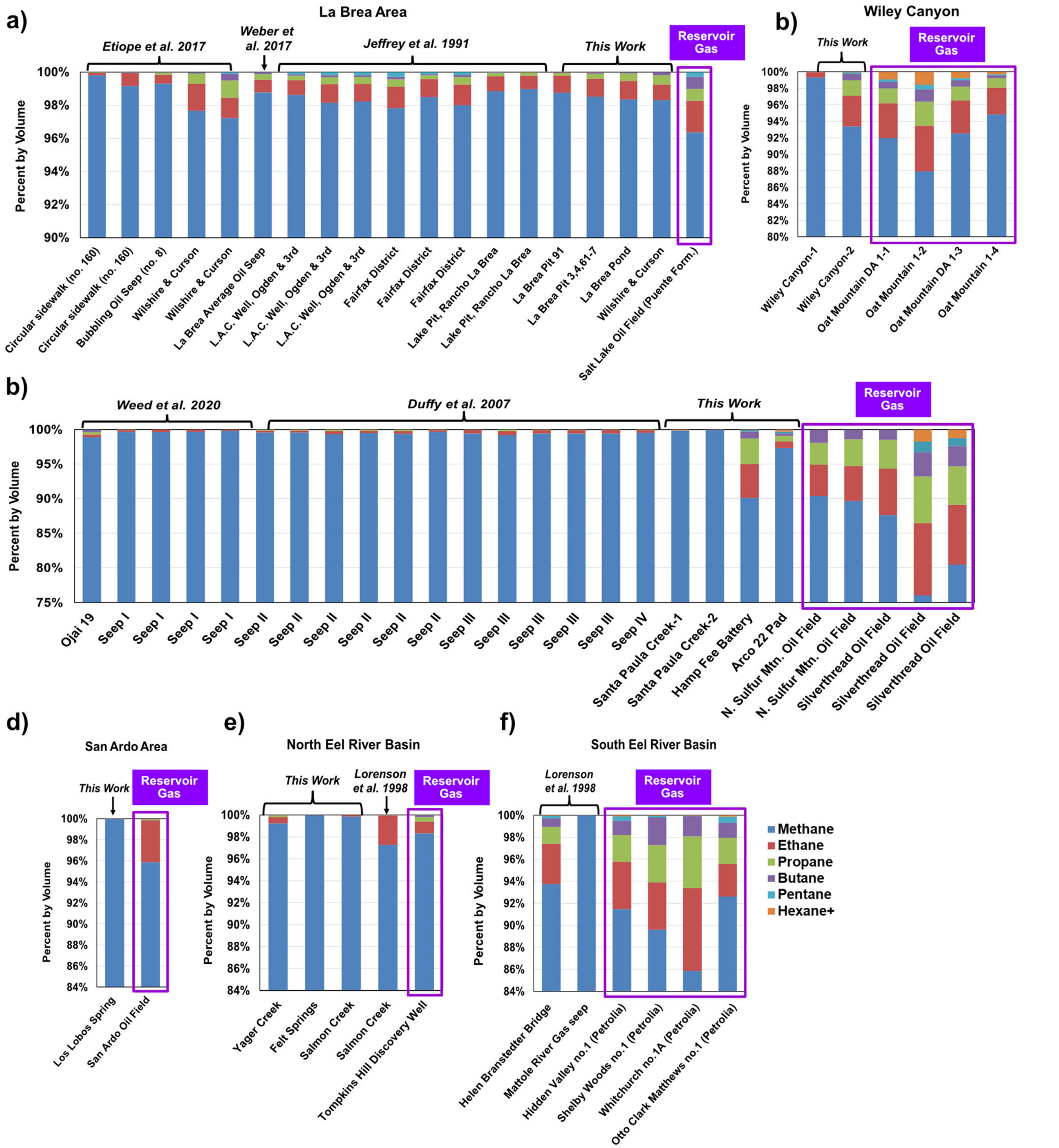

Fig. 5 | Hydrocarbon molecular composition (from methane to hexane+) of surface gas manifestations and local reservoir gases (designated by purple square). **a** La Brea area; **b** Wiley Canyon, **c** Upper Ojai Valley; **d** proximate to San Ardo Oil Field; **e** North Eel River Basin; and **f** South Eel River basin. Data sources for reservoir gases are reported in the "Data availability".

seepage. This model is intended to be used as a guide to discriminate the origin, natural vs anthropogenic, of free-phase gas entering the atmosphere, using ground-based or aerial measurements.

Before discussing the alterations, it is important to note that the stable C isotope composition of CH_4 ($\delta^{13}C$–C_1) is the one parameter that does not seem to be substantially modified during seepage (Figs. 2a and 4). This suggests that post-genetic processes that may alter the $^{13}C/^{12}C$ ratio of C_1 (aerobic and anaerobic microbial oxidation of C_1, mixing with microbial gases or thermogenic gases with a different $\delta^{13}C$–C_1 composition) are limited in their effect. Aerobic microbial oxidation is an important process acting on C_1 in the aerated vadose zone where gas can accumulate in soil and sediments[33,34]. However, in the case of seeps, aerobic microbial oxidation may occur in the aerated soil and vadose zone surrounding the seep, but not within the oxygen-free gas and water flow of the vent[35,36]. The lack of substantial $\delta^{13}C$–C_1 alteration has also been observed in seeps and mud volcanoes in other countries, and supports the assumption that gas migrates

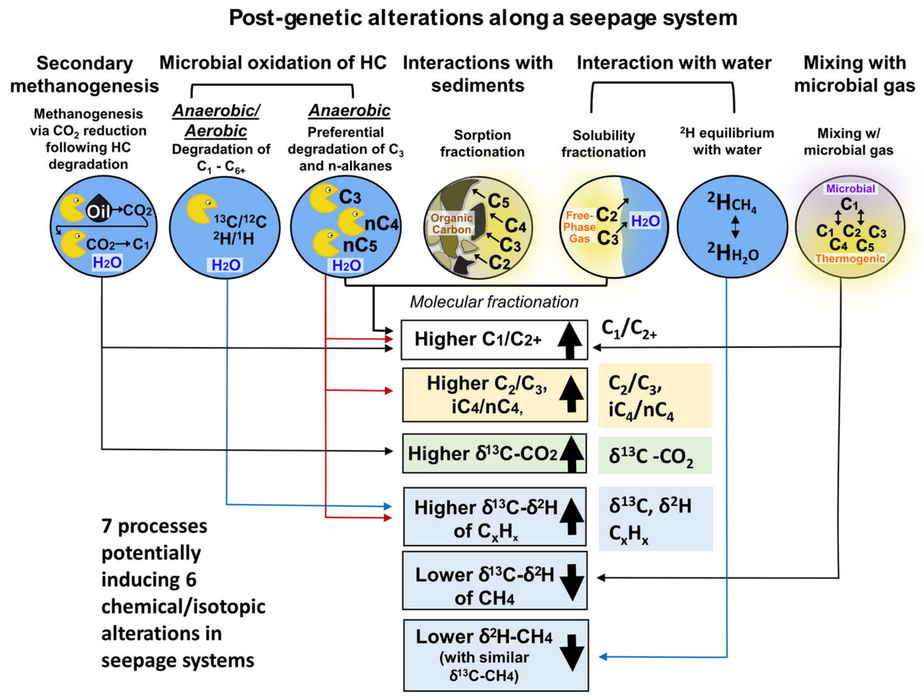

Fig. 6 | Conceptual model of post-genetic processes affecting seeping gas. The six types of alterations are typical of natural seeps (free-phase gas) stemming from advective migration from reservoirs.

dominantly by advection rather than diffusion (where the latter is associated with diffusive isotopic fractionation)[15,16,26,37]. Note that while evidence of secondary methanogenesis is commonly observed in surface manifestations (further discussion below), the $\delta^{13}C$–C_1 composition of secondary methane can be quite similar to that of the parent oil substrate. As a result, secondary microbial CH_4 may be indistinguishable from oil-associated thermogenic CH_4, and therefore would not substantially affect the $\delta^{13}C$–C_1 composition[28,38,39]. In contrast to $\delta^{13}C$–C_1, there were several molecular or isotopic parameters that frequently exhibited differences between surface gas manifestations and underlying reservoir gas. These are discussed below.

Increased C_1/C_{2+} ratio. The most common change in gas chemistry observed during natural seepage is an increase in the C_1/C_{2+} ratio relative to reservoir gas (Figs. 2b and 4). The common nature of this change may be attributable to the multiple processes (five out of the seven processes shown in Fig. 6) that can contribute to C_{2+} loss and/or C_1 gain. These include: the different solubility and adsorption properties of C_{2+} gases relative to methane, secondary methanogenesis, preferential anaerobic oxidation of C_3 and *n*-alkanes, and mixing with microbial gas (Fig. 6).

Changes in δ^2H–C_1. The stable H isotope composition of CH_4, δ^2H–C_1, in surface manifestations, can vary considerably (e.g., >±20‰) from reservoir gas, though the direction of this variation is not uniform. ^2H enrichment in the surface gas can be due to CH_4 oxidation, if associated with ^{13}C enrichment, as is the case of Hamp Fee Battery, Arco 22 Pad, and Los Lobos Spring; Supplementary Fig. 3. The more pronounced enrichment in δ^2H–C_1 relative to $\delta^{13}C$–C_1 likely reflects the larger effect of microbial oxidation on the δ^2H–C_1 vs $\delta^{13}C$–C_1 (where δ^2H increases by 8–14 per mil for every 1 per mil enrichment in $\delta^{13}C$[34,40]). ^2H depletion can be due to mixing with microbial gas (if associated with ^{13}C depletion, as observed in La Brea). Alternatively, ^2H depletion can result from prolonged interaction with groundwater, i.e., partial equilibration with water's hydrogen[41,42], likely due to longer residence time in aquifers. In this case, ^2H depletion is not associated with ^{13}C depletion, as observed in gas from the Salmon Creek surface manifestation. On a global scale, there is no significant difference between the median δ^2H–C_1 values for seeps vs reservoir gases (Supplementary Fig. 4). The similarity of these datasets may reflect the fact that post-genetic alteration of δ^2H–C_1 is not largely unidirectional, and therefore, does not impact the central tendency of global δ^2H–C_1 seep vs reservoir data. However, on a site-specific basis, differences between δ^2H–C_1 of seeps and underlying reservoir gases may be readily apparent, as observed for several California seeps sampled in this study.

Increased $\delta^{13}C$–CO_2. The enriched $\delta^{13}C$–CO_2 composition of many surface manifestations (e.g., in the La Brea area and Upper Ojai Valley, as well as globally; Supplementary Fig. 5 and Fig. 2b, respectively) is consistent with secondary methanogenesis by CO_2 reduction following hydrocarbon biodegradation, where the latter produces CO_2 as a reaction byproduct[43]. Secondary methanogenesis commonly occurs within shallow (<2000 m) biodegraded oil-bearing reservoirs[28], or during the transport of oil and associated fluids (including gases) from these shallow reservoirs to the surface. For this reason, positive $\delta^{13}C$–CO_2 values are a well-established characteristic of oil-associated natural gas seeps[15,16]. As an example, numerous seeps from La Brea exhibit more positive $\delta^{13}C$–CO_2 values than reservoir gas, which itself is already biodegraded and characterized by ^{13}C-enriched CO_2. Of interest, a handful of La Brea seepage samples exhibit a lower $\delta^{13}C$–CO_2 value relative to reservoir gas. This decrease in $\delta^{13}C$–CO_2 can result from microbial oxidation of C_1–C_{6+} (where ^{13}C-depleted CO_2 is a reaction byproduct). Overall, biodegraded reservoirs are not common in global natural gas production (Fig. 2; consequently, fugitive emissions mostly refer to gas produced in large, deeper, and non-biodegraded reservoirs with "normal" $\delta^{13}C$–CO_2 values[8,29].

Increased C_2/C_3 and iC_4/nC_4 ratios. Many surface manifestations exhibit increased C_2/C_3 and iC_4/nC_4 ratios relative to reservoir gas (Supplementary Tables 1–3), which is indicative of the preferential microbial degradation (anaerobic oxidation) of C_3 and *n*-alkanes[43]. This is a noted characteristic of La Brea seepage[9,30,44].

Increased $\delta^{13}C$–C_{3+}. The preferential degradation of C_{3+} alkanes results in ^{13}C-enrichment in these molecules. In Ojai Valley manifestations, we observed a large increase of $\delta^{13}C$–C_3 (i.e., >+25‰) relative to reservoir gas, as compared to $\delta^{13}C–C_1$ and $–C_2$ enrichment (i.e., >+5‰) (Supplementary Fig. 6).

Conditions when seeping gas is not substantially altered. Three surface gas expressions sampled in this study (the intersection of Wilshire Boulevard and Curson Avenue, Hamp Fee Battery, Wiley Canyon #2) exhibited a molecular and CH_4 isotopic composition similar to reservoir gas (i.e., negligible C_{2+} loss or change in $\delta^{13}C–C_1$ or $\delta^2H–C_1$ where these data were available for reservoir gas; Fig. 4 and Supplementary Fig. 3). Based on visual observation (e.g., vigorous bubbling) and/or free-phase gas concentration measurements in ambient air exceeding the level of explosivity, these manifestations were characterized by higher gas flow rates than other manifestations. The correlation between gas flux and the $C_1/(C_2 + C_3)$ ratio is known in seepage studies, especially for mud volcanoes[24,45]. Specifically, higher natural gas seepage flux is inversely related to post-genetic alteration; i.e., the higher the gas flux, the higher the gas velocity, and therefore, the less time available for gas–water interaction and molecular fractionation. All three of these large and very active manifestations are located proximate to noted faults (see "Discussion" in Supplementary Information), which may act as higher permeability migration pathways. As with many of the natural seeps sampled in this study, these three manifestations are also located proximate (<500 m) to active or historical oil wells, another potential source of reservoir gas. We posit that the similarities of natural seepage gas to reservoir gas suggest, at the most basic interpretation, a more direct migration pathway with higher-volume flow. However, Hamp Fee battery gas has a notably more positive $\delta^{13}C–CO_2$ value (+26.8‰) than local reservoir gases (+8.8 to 10.6‰), which suggests that the manifestation is natural. The $\delta^{13}C–CO_2$ value for Wiley Canyon #2 is also positive (+15.4‰), although $\delta^{13}C–CO_2$ data for underlying reservoir gases are not available, which prevents further interpretation. Concerning the manifestation of Wilshire Boulevard and Curson Avenue in Los Angeles, the oil well database of the Department of Conservation's Division of Oil, Gas, and Geothermal Resources of California does not report any well at that site, and the Los Angeles city officials considered the gas leak a natural seepage phenomenon[9].

Limitations of approach

There are certain limitations to this approach. First, it can be difficult to definitively identify the parent reservoir for surface gas manifestations. In the studied basins, gases present in vertically stacked reservoirs commonly originate from the same source rock and are compositionally similar[32]. Nevertheless, even in the case of multiple potential gas sources, broad compositional shifts away from reservoir gas (or gases) would likely be identifiable. Second, because this model relies on changes to gas chemistry during transport, there may be situations where natural gas seepage has migrated via a pathway with high flow and short transport time, and therefore is indistinguishable from reservoir gas. However, based on the global and local datasets presented herein, such situations are the exceptions rather than the rule. Third, this model does not apply to hydrocarbon gases dissolved in groundwater, which may undergo additional alteration that further modifies the chemical composition[22]. Finally, our interpretations are based on the scope of available data and known processes at this time.

Importantly, since the gas alteration processes take place in saturated underground sediments and rocks, our model applies equally to onshore and offshore systems. However, the application of the model to offshore systems should refer to gas sampled at the seabed, not at the seawater surface. During the transport of gas from the seabed to the atmosphere, additional alteration processes take place, such as dissolution (impacted by the differential solubility of alkanes), gas exchange between bubbles and seawater, and oxidation which require additional modeling if gas is examined at the sea surface. In addition, for deep submarine seeps (i.e., 300–400 m), little gas reaches the surface[46,47].

Potential model application

In focused ground-based surveys, the model, which utilizes the molecular and isotopic analyses shown in Fig. 6, can be directly applied by comparing the chemical composition of a discrete surface gas manifestation to that of unaltered reservoir gas (the source of most fugitive gas leaks). For example, identification of any one of the post-genetic alteration mechanisms reported in our model may allow for the likely differentiation of a natural seep vs fugitive emissions from abandoned and buried oil and gas wells.

On a larger scale, to attribute and quantify hydrocarbon gas emissions from seeps vs other sources in local or regional areas of gas-oil production, several aerial and ground-based monitoring approaches could be applied using the model outlined herein. Numerous studies have relied upon aerial or ground-based chemical and isotopic measurements to distinguish fugitive fossil emissions from other methane sources (i.e., landfills, wastewater treatment plants, dairy emissions)[10,12,18,19,48–52]. Chemical tracers utilized in these studies include: (a) the relative abundances of $C_1–C_5$ alkanes (including n- and i-alkanes) and CO_2 to quantify fossil gas emissions from natural gas in urban distribution systems and/or geologic seeps, as opposed to gases from landfills and dairies[52,53]; (b) $\delta^{13}C–C_1$ to characterize emissions from geologic seeps and the oil and gas sector (including pipelines and fossil fuels refining)[10,51]; (c) $\delta^2H–C_1$ to attribute methane emissions from coal mines and oil and gas extraction vs wastewater treatment[10,49]; and (d) $\delta^{13}C–CO_2$ to quantify CO_2 anthropogenic vs biogenic CO_2 sources[48]. However, where natural seeps were considered, these studies reported that they could not be distinguished from anthropogenic leaks of fossil gases[53]. Utilizing insights from this investigation (and outlined in Fig. 6), existing aerial and ground-based monitoring approaches may be successfully applied to evaluate the relative contributions of natural seeps vs fugitive leaks of reservoir gases and other sources of methane. In addition, with the dramatic increase in the range of vehicle-based monitoring platforms for source attribution and emission estimates in the last decade[54–58], such platforms may facilitate the monitoring of multiple molecular and isotopic parameters concurrently. More generally, the model can serve as a foundation for future work to better understand the relative contribution of natural seepage to local and regional methane emissions in thermogenic gas-bearing basins around the world.

Methods

Natural gas seeps in California were identified through a review of the Hodgson[59] compilation of "Onshore Oil & Gas Seeps in California," as well as numerous scientific publications and a global seeps dataset[9,30,44,60–63]. The reported locations of active natural seeps were visited in June and July of 2022. To evaluate where and if natural gas was surfacing, the area immediately overlying oil or water seepage was screened for methane (i.e., within 15 cm of the surface) using a Bascom-Turner Gas Explorer Detector, which utilizes dual catalytic methane sensors to provide measurements of combustible gases with 20 ppmv resolution. In some areas (e.g., Carpinteria State Beach), natural gas seepage was not located during the field visit, despite previous reports of its presence.

Eighteen surface gas manifestation samples were collected (Supplementary Table 1). Out of the 18 individual sampling locations, 12 surface manifestations produced oil in addition to gas, and water was observed in association with hydrocarbon seepage at six locations. Natural gas was concentrated prior to collection using a sampling shield constructed from a stainless steel bowl, which could be placed directly above the seepage location (Supplementary Fig. 2). Bowls of various sizes were utilized depending on the size and nature of surface gas expression. Each bowl was outfitted with a port on its apex, through which ¼-inch polyethylene tubing was threaded. Gas was allowed to be naturally purged via this port while natural gas accumulated within the bowl. The concentration of combustible gases purging through the tubing was checked at regular intervals using the Bascom Turner Gas Explorer Detector. When combustible gas concentrations were sufficiently high (e.g., greater than 5% *v*/*v*), a hand squeeze bulb was used to fill a pre-evacuated IsoBag™ from Isotech Laboratories (Isotech) with approximately 300 cm^3 of the collected gas. At select locations with lower relative gas production rates, IsoBags™ were filled at lower accumulated combustible gas concentrations (e.g., greater than 500 parts per million by volume (ppmv)). At a single seepage location (Los Lobos Spring), a dissolved gas sample was collected in an IsoFlask™ per the methods described in ref. 64. In addition to samples of surface manifestation gas, a sample of dissolved reservoir gas was collected using an IsoFlask™ container from a tank containing a mixture of produced water from multiple oil wells in the San Ardo oil field, and two samples of reservoir gas were collected in a 300 cubic centimeter double-ended steel gas cylinder directly from the bradenhead of oil wells in the Silverthread oil field.

Gas samples were analyzed for C_1–C_{6+} hydrocarbons and non-hydrocarbon gases (CO_2, Ar, O_2, and N_2) by Isotech Laboratories (Isotech), in Champaign, Illinois on a Shimadzu 2010 Gas Chromatograph (GC) equipped with both a thermal conductivity detector (TCD) and flame ionization detector (FID). Reported analytical errors of external standards are as follows: C_1–C_4: ±5%, C_5–C_{6+}: ±10%, N_2, CO_2, O_2, and Ar: ±10%. For samples where gas concentrations were sufficient, $\delta^{13}C$ and δ^2H of C1, and $\delta^{13}C$ of C_2, C_3, and CO_2 were analyzed via dual-inlet isotope ratio mass spectrometer (DI-IRMS) using a Finnigan MAT Delta S Isotope Ratio Mass Spectrometer ($\delta^{13}C$ of C_1, C_2, C_3, and CO_2) and a Finnigan Delta Plus XL isotope ratio mass spectrometer (δ^2H of C_1). $\delta^{13}C$ and δ^2H values are reported relative to VPDB and VSMOW, respectively, on a scale defined by two-point calibration of LSVEC and NBS19 reference standards ($\delta^{13}C$), and VSMOW2 and SLAP2 reference standards (δ^2H). Two-sigma errors for $\delta^{13}C$ and δ^2H measurements of gases are ±0.2‰ and ±7‰. For select samples where concentrations of C_1, C_2, C_3, or CO_2 were insufficient for isotopic analysis by DI-IRMS, samples were analyzed using gas chromatography combustion isotope ratio mass spectrometry (GC-C-IRMS) at Isotech for $\delta^{13}C$ and δ^2H with 2-sigma errors of ±0.6‰ and ±10‰, respectively. Gas analytical data were corrected for air contamination during sampling based on the atmospheric N_2/O_2 ratio.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Global reservoir gas geochemistry. We used the molecular and isotopic composition of conventional thermogenic reservoir gases from ref. 65. Global seep gas geochemistry: we used the molecular and isotopic composition of thermogenic natural gas seepage from the Organic Geochemistry Data from Frogi and Fluid Features Database (CGG[63], with license restrictions), with additional data from the following sources: Baciu et al.[15]; Etiope et al.[28]; Jeffrey et al.[30]; Tassi et al.[66]; Zheng et al.[67]. For both global datasets (reservoir and seep gas geochemistry), only samples with $\delta^{13}C$–C_1 values consistent with mature thermogenic gas, with limited or no mixing with microbial gas (>−50‰, based on natural gas genetic diagrams from Milkov and Etiope[29]) were utilized. California reservoir gas geochemistry: we used data from producing oil and gas wells obtained from this study and from Duffy et al.[60]; Jeffrey et al.[30]; Lorenson et al.[61]; and Termco Company[68] (Supplementary Table 3). California seep gas geochemistry: in addition to the original data acquired in this work, which are available in Supplementary Table 1, we used available data on the molecular and isotopic composition of California natural gas seepage from Duffy et al.[60]; Etiope et al.[9]; Jeffrey et al.[30]; Lorenson et al.; Weber et al.[44]; and Weed et al.[62] (Supplementary Table 2). The data that support the findings of this study are available in the Figshare data repository (https://doi.org/10.6084/m9.figshare.28012691)[69] and in the Supplementary Information.

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Author contributions
L.J.M., G.E., D.C.S., and M.A.E. designed the study. L.J.M. and G.E. interpreted all data and developed the manuscript. L.J.M. and M.A.E. executed field sampling. D.C.S. and M.A.E. contributed to data interpretation and manuscript preparation.

Competing interests
The authors declare no competing interests.

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